

ECE 332 — Homework 4: Generated Answers from HW4 Cheat Sheet

Each problem: state the goal equation $\rightarrow$ identify missing variables $\rightarrow$ solve each $\rightarrow$ plug back $\rightarrow$ box the answer.

Symbol reference (used throughout):

- l — antenna physical length (m)
- $\lambda = c/f$ — wavelength (m); $c = 3 \times 10^8$ m/s
- I_0 — peak current at antenna feed (A)
- R — distance from antenna to observation point (m)
- θ — angle from antenna axis (rad or deg)
- $\eta_0 = 120\pi \approx 377 \Omega$ — intrinsic impedance of free space
- $k = 2\pi/\lambda$ — wavenumber (rad/m)
- $S(R, \theta)$ — average power density radiated (W/m²)
- $F(\theta) = S(\theta)/S_{\max}$ — normalized radiation intensity (unitless)
- β — half-power beamwidth (rad or deg)
- Ω_p — pattern solid angle (sr)
- $D = 4\pi/\Omega_p$ — directivity (unitless, often dB)
- $A_e = \lambda^2 D / (4\pi)$ — effective aperture area (m²)
- A_p — physical cross-section area (m²)
- G_t, G_r — antenna gains (linear)
- P_t, P_r — transmitted / received power (W)
- $k_B = 1.38 \times 10^{-23}$ J/K — Boltzmann constant
- T_{sys} — system noise temperature (K)
- B — receiver bandwidth (Hz)

Key shortcuts: Hertzian dipole ($l/\lambda < 1/50$) $\Rightarrow D = 1.5$. Half-wave dipole ($l = \lambda/2$) $\Rightarrow D = 1.64$, $R_{\text{rad}} = 73.2 \Omega$. 3 dB $\Rightarrow$ linear gain of 2.

Problem 1 — Hertzian Dipole Power Density at 5 km

8 pts

Setup: $l = 1$ m, $f = 1$ MHz, $I_0 = 12$ A, $R = 5$ km, $\theta = 45^\circ$. Find $S(R, \theta)$.

DIPOLE TYPE BY LENGTH

Compute first $\lambda = c/f$, then ratio l/λ

l/λ	Type	Use formula
$< 1/50$	Hertzian (short)	HERTZIAN $S(R, \theta)$, $D = 1.5$
$= 1/4$	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
$= 1/2$	Half-wave	half-wave short-cut, $D = 1.64$, $R_{\text{rad}} = 73 \Omega$
$= 1$	Full-wave	ARB, $D = 2.41$, $R_{\text{rad}} = 199 \Omega$
other	Arbitrary	ARB formula

If $l/\lambda < 1/50$ STOP and use Hertzian. Do NOT plug small l into the arbitrary formula — wrong shape.

COMMON DIPOLE PARAMETERS

Type	l	D	D_{dB}	$R_{\text{rad}} (\Omega)$
Isotropic	N/A	1	0	—
Hertzian	$< \lambda/50$	1.5	1.76	$80\pi^2 (l/\lambda)^2$
Half-wave	$\lambda/2$	1.64	2.15	73.2
$\lambda/4$ monopole	$\lambda/4$ (gnd)	1.64	2.15	36.6
Full-wave	λ	2.41	3.82	199

Half-wave: $P_{\text{rad}} = 36.6 I_0^2$ W. $\lambda/4$ monopole (above ground): same patt. as half-wave but P_{rad} , R_{rad} halved.

HERTZIAN DIPOLE — $S(R, \theta)$

Far-field E,H phasors small $l \ll \lambda$
 $\tilde{E}_\theta = \frac{j I_0 k l \eta_0}{4\pi} \frac{e^{-jkR}}{R} \sin \theta, \quad \tilde{H}_\phi = \tilde{E}_\theta / \eta_0$
 Power density if $l/\lambda < 1/50$; far field
 $S(R, \theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \sin^2 \theta = S_{\text{max}} \sin^2 \theta \text{ (W/m}^2\text{)}$
 $S(R, \theta)$ — avg. pwr dens. at (R, θ) (W/m²)
 l — antenna length (m, the rod itself)
 I_0 — peak current in dipole (A)
 R — distance from antenna to obs. point (m)
 θ — angle from dipole axis (rad/°)
 $k = 2\pi/\lambda$ — wavenumber (rad/m)
 $\eta_0 = 120\pi \approx 377 \Omega$ — free-space impedance
 Max direction broadside $\theta_{\text{max}} = \pi/2$
 Total radiated power integrate over 4π
 $P_{\text{rad}} = \frac{\eta_0 \pi}{3} \left(\frac{I_0 l}{\lambda} \right)^2 \text{ (W)}$
 Directivity exact $D_{\text{Hertz}} = 1.5$ (1.76 dB)

Step 0 — Classify the dipole. From DIPOLE TYPE BY LENGTH, we need l/λ .

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{10^6} = 300 \text{ m}, \quad \frac{l}{\lambda} = \frac{1}{300} < \frac{1}{50} \Rightarrow \textbf{Hertzian}$$

Goal: find $S(R, \theta)$ using the Hertzian power density formula (from HERTZIAN DIPOLE box):

$$S(R, \theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \sin^2 \theta$$

Need: $\eta_0, k, I_0, l, R, \theta$.

Solve each unknown:

η_0 : constant, free-space impedance.

$$\eta_0 = 120\pi \Omega$$

k : wavenumber from wavelength.

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{300} \text{ rad/m}$$

I_0, l, R, θ : given.

$$I_0 = 12 \text{ A}, \quad l = 1 \text{ m}, \quad R = 5000 \text{ m}, \quad \theta = 45^\circ$$

Plug back in:

$$S = \frac{(120\pi)(2\pi/300)^2(12)^2(1)^2}{32\pi^2(5000)^2} \sin^2(45^\circ)$$

With $\sin^2(45^\circ) = 1/2$:

$$S = \frac{(120\pi)(4\pi^2/90000)(144)(1)}{32\pi^2(25 \times 10^6)} \cdot \frac{1}{2}$$

Simplify: numerator has $120\pi \cdot 4\pi^2 \cdot 144/90000 = 69120\pi^3/90000$, denominator $32\pi^2 \cdot 25 \times 10^6 = 8 \times 10^8 \pi^2$. Combine and multiply by $1/2$:

$$S(5 \text{ km}, 45^\circ) = 1.51 \times 10^{-9} \text{ W/m}^2 = 1.51 \text{ nW/m}^2$$

Sanity: 12 A at 1 MHz through 1 m radiates 1.5 nW/m^2 at 5 km. The $1/R^2$ falloff dominates — most power scatters into space, only a tiny fraction reaches any one point.

Problem 2 — Beam Parameters from $F(\theta) = e^{-20\theta^2}$ **15 pts****Setup:** $F(\theta) = \exp(-20\theta^2)$ for $0 \leq \theta \leq \pi$ (θ in radians). Find (a) HPBW β , (b) Ω_p , (c) D .**BEAM PARAMETERS — β , Ω_p , D** **Step 1 — Normalize** *always start here*

$$F(\theta) = S(\theta)/S_{\max} \text{ (unitless)}$$

Step 2 — HPBW β *half-power beamwidth, full angular width at -3 dB*Set $F(\theta) = 0.5$, solve for $\theta \rightarrow \theta_{1/2}$

$$\beta = 2\theta_{1/2} \text{ (symmetric beams, rad or } ^\circ)$$

Gaussian:

$$F(\theta_{1/2}) = e^{-a\theta_{1/2}^2} = 0.5 \Rightarrow \theta_{1/2} = \sqrt{\ln 2/a}$$

Step 3 — Pattern solid angle Ω_p *units: sr*

$$\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta \, d\theta \, d\phi$$

*No ϕ dep.: $\Omega_p = 2\pi \int_0^\pi F(\theta) \sin \theta \, d\theta$.**TI-36X Pro: [2nd] [d/dx] for $\int_{\square}^{\square}$, set RAD mode, multiply result by 2π .**Narrow-beam Gaussian shortcut ($\sin \theta \approx \theta$, $\rightarrow \infty$):*

$$\Omega_p \approx \frac{\pi}{a} = \frac{\pi \theta_{1/2}^2}{\ln 2} \quad (F = e^{-a\theta^2})$$

Step 4 — Directivity D

$$D = \frac{4\pi}{\Omega_p} \text{ (unitless)} \quad D_{\text{dB}} = 10 \log_{10} D$$

*Alt. with two orthogonal HPBWs: $D \approx 4\pi / (\beta_{xz}\beta_{yz})$.***Step 5 — Gain (if asked) ξ** *rad. eff., $0 < \xi \leq 1$*

$$G = \xi D \quad G_{\text{dB}} = 10 \log_{10} G$$

*Lossless / ideal antenna $\Rightarrow G = D$.***(a) Half-power beamwidth β (5 pts)****Goal:** find β using the HPBW procedure (BEAM PARAMETERS Step 2):Set $F(\theta) = 0.5$, solve for $\theta \rightarrow \theta_{1/2}$

$$\beta = 2\theta_{1/2} \text{ (symmetric beam)}$$

Need: $\theta_{1/2}$, found by setting our specific $F(\theta) = 0.5$.**Solve $\theta_{1/2}$:** plug $F = \exp(-20\theta^2)$:

$$\exp(-20\theta_{1/2}^2) = 0.5 \Rightarrow -20\theta_{1/2}^2 = \ln(0.5) = -0.693$$

$$\theta_{1/2}^2 = \frac{0.693}{20} = 0.03466, \quad \theta_{1/2} = \sqrt{0.03466} = 0.186 \text{ rad}$$

That's the half-angle from peak to one -3 dB point. Since the beam is symmetric about $\theta = 0$, the *full* half-power beamwidth is twice that:

$$\beta = 2\theta_{1/2} = 0.372 \text{ rad} = 21.31^\circ$$

(b) Pattern solid angle Ω_p (5 pts)

Goal: find Ω_p using the integral definition (BEAM PARAMETERS Step 3):

$$\Omega_p = \iint_{4\pi} F(\theta) \sin \theta \, d\theta \, d\phi$$

Need: evaluate this integral with our specific $F(\theta) = e^{-20\theta^2}$.

Set up the integral: F has no ϕ dependence, so $\int_0^{2\pi} d\phi = 2\pi$:

$$\Omega_p = 2\pi \int_0^\pi e^{-20\theta^2} \sin \theta \, d\theta$$

Solve numerically on TI-36X Pro (no closed form):

1. [mode] $\rightarrow$ **RAD**
2. [2nd] [d/dx] for $\int_{\square}^{\square}$ template
3. Lower limit: 0; Upper limit: $[\pi]$
4. Integrand: $e^{(-20 \times x^2)} \times \sin(x)$
5. Variable: x ; [enter] $\rightarrow$ returns ≈ 0.024792
6. Multiply: $\times [\pi] \times 2$ [enter] $\rightarrow \approx 0.15578$

$$\Omega_p \approx 0.1558 \text{ sr}$$

Keep the unrounded value for part (c). Textbook rounds to 0.156 first and gets $D = 80.55$; using full precision gives $D \approx 80.67$.

(c) Directivity D (5 pts)

Goal: find D using the standard formula (BEAM PARAMETERS Step 4):

$$D = \frac{4\pi}{\Omega_p}$$

Need: Ω_p (from part b).

Plug in: $\Omega_p = 0.15578 \text{ sr}$:

$$D = \frac{4\pi}{0.15578} = \frac{12.566}{0.15578}$$

$$D = 80.67$$

In dB: $D_{dB} = 10 \log_{10}(80.67) = 19.07 \text{ dB}$.

Sanity: $D \approx 80$ is highly directive (much narrower than half-wave's $D = 1.64$). The narrow Gaussian-like beam ($\beta \approx 21^\circ$) concentrates radiation into a small solid angle, so directivity is large.

Problem 3 — $\lambda/4$ Dipole Radiation Pattern

15 pts

Setup: Dipole of length $l = \lambda/4$. Find (a) $\theta_{\max}$, (b) $S_{\max}$, (c) plot $F(\theta)$.

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Power density at R *any* l

$$S(\theta) = \frac{15I_0^2}{\pi R^2} \left[\frac{\cos\left(\frac{\pi l}{\lambda} \cos \theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin \theta} \right]^2$$

At $\theta = 0, \pi$: bracket is 0/0. *L'Hôpital* $\Rightarrow S = 0$ (no axial radiation). TI-36X Pro alt: evaluate bracket at $\theta = 0.001$ rad $\rightarrow$ tiny $\rightarrow 0$.

Normalized pattern *dimensionless* $F(\theta) = S(\theta)/S_{\max}$

Quarter-wave $l = \lambda/4$ $\pi l/\lambda = \pi/4$, $\cos(\pi/4) = \sqrt{2}/2$

$$\frac{S(\theta)}{S_0} = \left[\frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2, \quad S_0 = \frac{15I_0^2}{\pi R^2}$$

$$S_{\max} = S(\pi/2) = S_0 \left(1 - \frac{\sqrt{2}}{2}\right)^2 \approx 0.0858 S_0$$

Half-wave $l = \lambda/2$ $\pi l/\lambda = \pi/2$, *shortcut form*

$$S(\theta) = \frac{15I_0^2}{\pi R^2} \frac{\cos^2\left(\frac{\pi}{2} \cos \theta\right)}{\sin^2 \theta}$$

$D = 1.64$, $R_{\text{rad}} = 73 \Omega$, $\theta_{\max} = \pi/2$.

COMMON DIPOLE PARAMETERS

Type	l	D	D_{dB}	$R_{\text{rad}} (\Omega)$
Isotropic	N/A	1	0	—
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Half-wave: $P_{\text{rad}} = 36.6 I_0^2 W$. $\lambda/4$ monopole (above ground): same patt. as half-wave but P_{rad} , R_{rad} halved.

COMMON ANGLES (deg $\leftrightarrow$ rad, trig values)

$^\circ$	rad	$\sin \theta$	$\cos \theta$	$\sin^2 \theta$	$\cos^2 \theta$
0	0	0	1	0	1
30	$\pi/6$	1/2	$\sqrt{3}/2$	1/4	3/4
45	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	1/2	1/2
60	$\pi/3$	$\sqrt{3}/2$	1/2	3/4	1/4
90	$\pi/2$	1	0	1	0
180	π	0	-1	0	1

Convert: $\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \pi/180$. Antenna formulas usually take θ in rad.

Common setup for all 3 parts.

Goal: starting point for parts (a)-(c) is the arbitrary-length dipole power density (from ARBITRARY-LENGTH DIPOLE box):

$$S(\theta) = \frac{15I_0^2}{\pi R^2} \left[\frac{\cos\left(\frac{\pi l}{\lambda} \cos \theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin \theta} \right]^2$$

Need: plug in $l = \lambda/4$ to simplify.

Substitute $l = \lambda/4 \Rightarrow \pi l/\lambda = \pi/4$:

$$S(\theta) = \frac{15I_0^2}{\pi R^2} \left[\frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \cos(\pi/4)}{\sin \theta} \right]^2$$

From COMMON ANGLES table: $\cos(\pi/4) = \sqrt{2}/2 = 0.7071$.

$$S(\theta) = S_0 \left[\frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2, \quad S_0 \equiv \frac{15I_0^2}{\pi R^2}$$

(a) Direction of maximum radiation $\theta_{\max}$ (5 pts)

Goal: find $\theta_{\max}$ where $S(\theta)$ is maximum. **Need:** know the shape of $S(\theta)$ over $\theta \in [0, \pi]$.
 Three-step argument: endpoints, symmetry, sampling.

Step (i) — endpoints via L'Hôpital. Define $f(\theta) = \cos(\frac{\pi}{4} \cos \theta) - \frac{\sqrt{2}}{2}$ and $g(\theta) = \sin \theta$. At $\theta = 0$:

$$f(0) = \cos(\pi/4 \cdot 1) - \sqrt{2}/2 = \sqrt{2}/2 - \sqrt{2}/2 = 0$$

$$g(0) = \sin(0) = 0 \Rightarrow 0/0 \text{ form, apply L'Hôpital.}$$

Derivatives (chain rule for f):

$$f'(\theta) = \frac{\pi}{4} \sin(\frac{\pi}{4} \cos \theta) \sin \theta, \quad g'(\theta) = \cos \theta$$

$$\lim_{\theta \rightarrow 0} \frac{f'(\theta)}{g'(\theta)} = \frac{(\pi/4) \sin(\pi/4) \cdot 0}{1} = 0$$

So $S(0)/S_0 = 0^2 = 0$. Same algebra at $\theta = \pi$: $S(\pi) = 0$. **Peak is in $(0, \pi)$.**

Step (ii) — symmetry. Substitute $\theta \rightarrow \pi - \theta$: $\cos(\pi - \theta) = -\cos \theta$, but $\cos$ is even so $\cos((\pi/4) \cos(\pi - \theta)) = \cos((\pi/4) \cos \theta)$. And $\sin(\pi - \theta) = \sin \theta$. So $S(\theta) = S(\pi - \theta)$. **Pattern mirrors about $\theta = \pi/2$.**

Step (iii) — sample to confirm single peak.

At $\theta = 30^\circ$: $\cos(30^\circ) = 0.866$, $(\pi/4)(0.866) = 0.680$, $\cos(0.680) = 0.777$, $0.777 - 0.707 = 0.070$, $\sin(30^\circ) = 0.500$, $0.070/0.500 = 0.140$, $(0.140)^2 = \mathbf{0.020}$.

At $\theta = 60^\circ$: $\cos(60^\circ) = 0.500$, $(\pi/4)(0.500) = 0.393$, $\cos(0.393) = 0.924$, $0.924 - 0.707 = 0.217$, $\sin(60^\circ) = 0.866$, $0.217/0.866 = 0.250$, $(0.250)^2 = \mathbf{0.063}$.

At $\theta = 90^\circ$: $\cos(90^\circ) = 0$, $(\pi/4)(0) = 0$, $\cos(0) = 1$, $1 - 0.707 = 0.293$, $\sin(90^\circ) = 1$, $0.293/1 = 0.293$, $(0.293)^2 = \mathbf{0.0858}$.

θ	30°	60°	90°
S/S_0	0.020	0.063	0.0858

Monotone rise to 90° , then symmetric fall (by step (ii)). Single peak at the center.

$$\theta_{\max} = \pi/2 = 90^\circ \text{ (broadside)}$$

(b) Expression for $S_{\max}$ (5 pts)

Goal: find $S_{\max}$ by evaluating S at the maximum direction:

$$S_{\max} = S(\theta_{\max})$$

Need: $\theta_{\max}$ (from part a: $\theta_{\max} = \pi/2$).

Plug $\theta = \pi/2$ into $S(\theta)$ step by step:

Inner cos: $\cos(\pi/2) = 0$, so $\frac{\pi}{4} \cos(\pi/2) = 0$.

Outer cos: $\cos(0) = 1$.

Denominator: $\sin(\pi/2) = 1$.

Bracket: $(1 - \sqrt{2}/2)/1 = 1 - \sqrt{2}/2 = 1 - 0.7071 = 0.2929$.

Square: $(0.2929)^2 = 0.0858$.

Multiply by S_0 :

$$S_{\max} = 0.0858 \cdot \frac{15I_0^2}{\pi R^2} \approx \frac{1.287 I_0^2}{\pi R^2}$$

(c) Normalized pattern $F(\theta) = S(\theta)/S_{\max}$ (5 pts)

Goal: find $F(\theta)$ by dividing $S(\theta)$ by $S_{\max}$, then plot. **Need:** $S(\theta)$ and $S_{\max}$ (both from above).

Compute the normalization coefficient: $1/0.0858 = 11.66$.

Form $F(\theta)$:

$$F(\theta) = 11.66 \left[\frac{\cos\left(\frac{\pi}{4} \cos \theta\right) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2$$

Plot: bell-shaped curve peaking at $\theta = \pi/2$ with $F = 1$, dropping to 0 at $\theta = 0$ and π . Symmetric about $\theta = \pi/2$. Similar shape to half-wave dipole pattern but slightly narrower since the dipole is shorter.

Problem 4 — Effective Area vs. Physical Cross-section

5 pts

Setup: Half-wave dipole at $f = 100$ MHz, wire diameter $d = 2$ cm. Find A_e and compare to A_p .

EFFECTIVE AREA A_e		COMMON DIPOLE PARAMETERS				
Definition <i>antenna's RX cross section (m^2)</i> $A_e = \frac{\lambda^2 D}{4\pi} \text{ (m}^2\text{)}$		Type	l	D	D_{dB}	$R_{rad} \text{ (}\Omega\text{)}$
Physical cross section <i>wire dipole, diameter d</i> $A_p = l d = \frac{\lambda}{2} d \text{ (half-wave)}$		Isotropic	N/A	1	0	—
Inverse: D from A_e $D = \frac{4\pi A_e}{\lambda^2} \text{ (unitless)}$		Hertzian	$< \lambda/50$	1.5	1.76	$80\pi^2 (l/\lambda)^2$
Hertzian special case $D=1.5$ $A_e^{\text{Hertz}} = \frac{3\lambda^2}{8\pi} \approx 0.119\lambda^2 \text{ (m}^2\text{)}$		Half-wave	$\lambda/2$	1.64	2.15	73.2
<i>Thin wire: $A_e \gg A_p$. Antenna captures field over $\sim \lambda$.</i>		$\lambda/4$ monopole	$\lambda/4$ (gnd)	1.64	2.15	36.6
		Full-wave	λ	2.41	3.82	199
		<i>Half-wave: $P_{rad} = 36.6 I_0^2 W$. $\lambda/4$ monopole (above ground): same patt. as half-wave but P_{rad}, R_{rad} halved.</i>				

Part 1 — Effective area A_e

Goal: find A_e using the standard formula (from EFFECTIVE AREA box):

$$A_e = \frac{\lambda^2 D}{4\pi}$$

Need: λ and D .

Solve each unknown:

λ : use $\lambda = c/f$.

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{10^8} = 3 \text{ m}$$

D : look up half-wave in COMMON DIPOLE PARAMETERS table.

$$D_{\text{half-wave}} = 1.64$$

Plug back in:

$$A_e = \frac{(3)^2(1.64)}{4\pi} = \frac{9 \cdot 1.64}{4\pi} = \frac{14.76}{12.566}$$

$$A_e = 1.17 \text{ m}^2$$

Part 2 — Physical cross-section A_p

Goal: find A_p using the wire-dipole formula:

$$A_p = l \cdot d$$

Need: l and d .

Solve each unknown:

l : half-wave means $l = \lambda/2$ (from COMMON DIPOLE PARAMS table).

$$l = \frac{\lambda}{2} = \frac{3}{2} = 1.5 \text{ m}$$

d : given.

$$d = 2 \text{ cm} = 0.02 \text{ m}$$

Plug back in:

$$A_p = (1.5)(0.02) = 0.03 \text{ m}^2$$

Part 3 — Ratio

$$\frac{A_e}{A_p} = \frac{1.17}{0.03} = \boxed{39.2}$$

Sanity: the antenna's "catching area" is $39\times$ its physical wire cross-section. A thin wire interacts with the field over a region $\sim \lambda$ around itself, so it captures much more energy than its tiny wire would suggest.

Problem 5 — Friis Transmission, TV Broadcast

5 pts

Setup: Half-wave dipole TX, $P_t = 1$ kW at $f = 50$ MHz. RX antenna with $G_r = 3$ dB, distance $R = 30$ km. Find P_r .

FRIIS TRANSMISSION FORMULA

Far-field link budget P_t tx, P_r rx, range R

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t \quad (\text{W})$$

G_t, G_r are linear gains; $\lambda = c/f$; R in m; P_t, P_r in W.
Solve for R max range

$$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}} \quad (\text{m})$$

Equivalent form using A_e recv. area form

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.

General form (off-axis) when patterns NOT peak-aligned

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$$

F_t, F_r normalized patterns at the link directions; $F = 1$ when aligned.

Recipe 1. $\lambda = c/f$. 2. Convert dB gains to linear: $G = 10^{G_{dB}/10}$.

3. If TX is “half-wave dipole” use $G_t = 1.64$. 4. Plug into Friis.

Use linear G , not dB. Convert FIRST.

dB ↔ LINEAR

$$G_{dB} = 10 \log_{10} G_{lin} \quad G_{lin} = 10^{G_{dB}/10}$$

dB		linear	
0	1	10	10
3	2	13	20
6	4	20	100
7	5	30	1000
-3	0.5	-10	0.1

Memorize: 3 dB $\Leftrightarrow \times 2$, 10 dB $\Leftrightarrow \times 10$. Add dB = multiply linear.

For E-field / amplitude use 20 log not 10 log (power $\propto E^2$).

Goal: find P_r using the Friis transmission formula (from FRIIS TRANSMISSION FORMULA box):

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t$$

Need: G_t, G_r (both linear), λ, R, P_t .

Solve each unknown:

λ : $\lambda = c/f$.

$$\lambda = \frac{3 \times 10^8}{5 \times 10^7} = 6 \text{ m}$$

G_t : TX is half-wave dipole, COMMON DIPOLE PARAMS gives $D = 1.64$, and for lossless antenna $G = D$.

$$G_t = 1.64$$

G_r : given as 3 dB. Convert to linear via dB $\rightarrow$ LINEAR table: 3 dB = $\times 2$.

$$G_r = 10^{3/10} = 2$$

R, P_t : given.

$$R = 30 \text{ km} = 3 \times 10^4 \text{ m}, \quad P_t = 10^3 \text{ W}$$

Plug back in: compute the path-loss factor first.

$$\frac{\lambda}{4\pi R} = \frac{6}{4\pi(3 \times 10^4)} = \frac{6}{3.77 \times 10^5} = 1.59 \times 10^{-5}$$

$$\left(\frac{\lambda}{4\pi R} \right)^2 = (1.59 \times 10^{-5})^2 = 2.53 \times 10^{-10}$$

$$P_r = (1.64)(2)(2.53 \times 10^{-10})(10^3) = (3.28)(2.53 \times 10^{-7})$$

$$P_r = 8.3 \times 10^{-7} \text{ W} = 0.83 \mu\text{W}$$

Sanity: 1 kW in, 0.83 μ W out at 30 km. A 90 dB drop, dominated by $1/R^2$ path loss. Plenty for a TV receiver (sensitivity $\sim 10^{-13}$ W).

Problem 6 — Communication Link with SNR

15 pts (new)

Setup: $G_t = 20$ dB, $G_r = 23$ dB, $R = 20$ km, $P_t = 10$ W, $f = 6$ GHz. (c) $T_{\text{sys}} = 1000$ K, $B = 20$ MHz.

FRIIS TRANSMISSION FORMULA

Far-field link budget P_t tx, P_r rx, range R

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t \quad (\text{W})$$

 G_t, G_r are linear gains; $\lambda = c/f$; R in m; P_t, P_r in W.Solve for R max range

$$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}} \quad (\text{m})$$

Equivalent form using A_e recv. area form

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.

General form (off-axis) when patterns NOT peak-aligned

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$$

 F_t, F_r normalized patterns at the link directions; $F = 1$ when aligned.Recipe 1. $\lambda = c/f$. 2. Convert dB gains to linear: $G = 10^{G_{\text{dB}}/10}$.3. If TX is "half-wave dipole" use $G_t = 1.64$. 4. Plug into Friis.Use linear G , not dB. Convert FIRST.

NOISE & SNR

Noise power thermal noise into matched receiver

$$P_N = k_B T_{\text{sys}} B \quad (\text{W})$$

 $k_B = 1.38 \times 10^{-23}$ J/K; T_{sys} system noise temp (K); B receiver bandwidth (Hz).

Signal-to-noise ratio

$$\text{SNR} = \frac{P_{\text{rec}}}{P_N} \quad (\text{unitless}) \quad \text{SNR}_{\text{dB}} = 10 \log_{10} \frac{P_{\text{rec}}}{P_N}$$

Typical comm. link needs SNR > 10 dB for usable signal.

dB ↔ LINEAR

$$G_{\text{dB}} = 10 \log_{10} G_{\text{lin}} \quad G_{\text{lin}} = 10^{G_{\text{dB}}/10}$$

	dB	linear	dB	linear
	0	1	10	10
	3	2	13	20
	6	4	20	100
	7	5	30	1000
	-3	0.5	-10	0.1

Memorize: 3 dB $\Leftrightarrow \times 2$, 10 dB $\Leftrightarrow \times 10$. Add dB = multiply linear.For E-field / amplitude use $20 \log$ not $10 \log$ (power $\propto E^2$).

Shared setup: convert dB to linear and compute λ .

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{6 \times 10^9} = 0.05 \text{ m}$$

$$G_t = 10^{20/10} = 100, \quad G_r = 10^{23/10} = 199.5$$

(a) Power density at RX antenna S_r (5 pts)

Goal: find S_r using power-spreading-over-sphere (with TX gain factored in):

$$S_r = \frac{G_t P_t}{4\pi R^2}$$

Need: G_t, P_t, R .

Plug in:

$$S_r = \frac{(100)(10)}{4\pi(2 \times 10^4)^2} = \frac{1000}{4\pi(4 \times 10^8)} = \frac{1000}{5.027 \times 10^9}$$

$$S_r = 1.989 \times 10^{-7} \text{ W/m}^2 \approx 0.199 \text{ } \mu\text{W/m}^2$$

(b) Received power P_r (5 pts)**Goal:** find P_r using Friis (from FRIIS TRANSMISSION box):

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t$$

Need: G_t , G_r , λ , R , P_t (all already converted).**Plug in:**

$$\frac{\lambda}{4\pi R} = \frac{0.05}{4\pi(2 \times 10^4)} = 1.989 \times 10^{-7}$$

$$\left(\frac{\lambda}{4\pi R} \right)^2 = 3.957 \times 10^{-14}$$

$$P_r = (100)(199.5)(3.957 \times 10^{-14})(10) = 7.894 \times 10^{-9}$$

$$\boxed{P_r = 7.89 \text{ nW}}$$

Cross-check via effective area: $A_{e,r} = G_r \lambda^2 / (4\pi) = 199.5(0.0025)/(4\pi) = 0.0397 \text{ m}^2$, then $P_r = S_r A_{e,r} = (1.989 \times 10^{-7})(0.0397) = 7.89 \times 10^{-9} \text{ W}$ ✓.

(c) Signal-to-noise ratio in dB (5 pts)**Goal:** find SNR in dB. From NOISE & SNR box, two steps:

$$P_N = k_B T_{\text{sys}} B \quad \text{then} \quad \text{SNR}_{\text{dB}} = 10 \log_{10} \frac{P_r}{P_N}$$

Need: P_N first ($\rightarrow$ need k_B , T_{sys} , B), then P_r (from part b).**Solve each unknown:** k_B : Boltzmann constant.

$$k_B = 1.38 \times 10^{-23} \text{ J/K}$$

 T_{sys} , B : given.

$$T_{\text{sys}} = 1000 \text{ K}, \quad B = 20 \text{ MHz} = 2 \times 10^7 \text{ Hz}$$

Compute P_N :

$$P_N = (1.38 \times 10^{-23})(1000)(2 \times 10^7) = 2.76 \times 10^{-13} \text{ W}$$

Compute SNR:

$$\text{SNR}_{\text{lin}} = \frac{P_r}{P_N} = \frac{7.89 \times 10^{-9}}{2.76 \times 10^{-13}} = 2.86 \times 10^4$$

$$\boxed{\text{SNR}_{\text{dB}} = 10 \log_{10}(2.86 \times 10^4) = 44.6 \text{ dB}}$$

Sanity: $>40 \text{ dB}$ SNR is a very clean link. Plenty for digital comm.

Problem 7 — Circular Aperture Antenna Scaling

15 pts (new)

Setup: Circular aperture, circular beam $\beta = 3^\circ$ at $f = 20$ GHz. (a) D in dB. (b) New D and β if A_p doubled. (c) New D and β if frequency doubled (same A_p).

APERTURE ANTENNAS (horn, dish)

Directivity from aperture area *uniform illumination*

$$D \approx \frac{4\pi A_p}{\lambda^2} \text{ (unitless)} \Rightarrow A_e \approx A_p$$

Directivity from beamwidths *rectangular beam*

$$D \approx \frac{4\pi}{\beta_{xz}\beta_{yz}} \text{ (use rad)}$$

Circular beam ($\beta_{xz} = \beta_{yz} = \beta$): $D \approx 4\pi/\beta^2$.Half-power beamwidth *uniform rect. illumination*

$$\beta_{xz} \approx 0.88 \frac{\lambda}{\ell_x} \text{ (rad)} \quad \beta_{yz} \approx 0.88 \frac{\lambda}{\ell_y}$$

Scaling: $D \propto A_p/\lambda^2$; $\beta \propto \lambda/\sqrt{A_p}$. Double area $\Rightarrow D \times 2$, $\beta/\sqrt{2}$. Double f ($\lambda/2$) $\Rightarrow D \times 4$, $\beta/2$.

dB $\leftrightarrow$ LINEAR

$$G_{\text{dB}} = 10 \log_{10} G_{\text{lin}} \quad G_{\text{lin}} = 10^{G_{\text{dB}}/10}$$

dB	linear	dB	linear
0	1	10	10
3	2	13	20
6	4	20	100
7	5	30	1000
-3	0.5	-10	0.1

Memorize: 3 dB $\Leftrightarrow \times 2$, 10 dB $\Leftrightarrow \times 10$. Add dB = multiply linear.

For E-field / amplitude use 20 log not 10 log (power $\propto E^2$).

(a) Directivity in dB (5 pts)

Goal: find D using the circular-beam formula (from APERTURE ANTENNAS box):

$$D = \frac{4\pi}{\beta^2} \quad (\beta \text{ in radians})$$

Need: β in radians (given as degrees).

Convert β to radians:

$$\beta = 3^\circ \cdot \frac{\pi}{180} = \frac{\pi}{60} = 0.05236 \text{ rad}$$

Plug in:

$$D = \frac{4\pi}{(0.05236)^2} = \frac{12.566}{0.002742} = 4584$$

$$D = 10 \log_{10}(4584) = 36.6 \text{ dB}$$

(b) Area doubled $A_p \rightarrow 2A_p$ (5 pts)

Goal: find new D and β using the scaling rules (from APERTURE ANTENNAS box coral note):

$$D \propto A_p \Rightarrow D_{\text{new}} = 2D_{\text{old}}$$

$$\beta \propto 1/\sqrt{A_p} \Rightarrow \beta_{\text{new}} = \beta_{\text{old}}/\sqrt{2}$$

Need: D_{old} and β_{old} (from part a).

Plug in:

$$D_{\text{new}} = 2(4584) = 9168 \Rightarrow D_{\text{new}} = 39.6 \text{ dB}$$

$$\beta_{\text{new}} = 3^\circ/\sqrt{2} = 3^\circ/1.414 = 2.12^\circ$$

(c) Frequency doubled $f \rightarrow 2f$ ($\lambda \rightarrow \lambda/2$), same A_p (5 pts)

Goal: find new D and β using scaling: from $D = 4\pi A_p/\lambda^2$ we have $D \propto 1/\lambda^2$. Halving λ quadruples D . And $\beta \propto \lambda$, so halving λ halves β . **Need:** D_{old} and β_{old} (from part a).

Plug in:

$$D_{\text{new}} = 4D_{\text{old}} = 4(4584) = 18336 \Rightarrow \boxed{D_{\text{new}} = 42.6 \text{ dB}}$$

$$\beta_{\text{new}} = \beta_{\text{old}}/2 = 3^\circ/2 = \boxed{1.5^\circ}$$

Sanity: bigger area or shorter wavelength both sharpen the beam. Doubling area: +3 dB ($\times 2$). Doubling frequency: +6 dB ($\times 4$) because both $D \propto A_p/\lambda^2$ and $1/\beta^2 \propto D$.

Problem 8 — 5-Element Linear Array, $d = 3\lambda/4$

(new)

Setup: Broadside linear array, $N = 5$ elements, equal-phase, uniform amplitude, spacing $d = 3\lambda/4$. Find (a) normalized array factor, (b) HPBW.

LINEAR ANTENNA ARRAY

Setup N identical elements, spacing d along axis

element pos $z_i = id$, $i = 0, \dots, N-1$; $A_i = a_i e^{j\psi_i}$

Pattern multiplication

$$S(R, \theta, \phi) = S_e(R, \theta, \phi) F_a(\theta)$$

Array factor $F_a(\theta) = \left| \sum A_i e^{jkid \cos \theta} \right|^2$

$$F_a(\theta) = \left| \sum_{i=0}^{N-1} A_i e^{jkid \cos \theta} \right|^2$$

Uniform $A_i = 1$, **equal phase** *closed form*

$$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}, \quad \gamma = kd \cos \theta$$

Normalize: F_a/N^2 . Peak N^2 at $\gamma = 0$ ($\cos \theta = 0$, broadside).

HPBW (broadside, uniform) *array length* $L = (N-1)d$
 $\beta \approx 0.88 \frac{\lambda}{L}$ (rad)

(a) Normalized array factor $F_a^{\text{norm}}(\theta)$ (5 pts)

Goal: find $F_a^{\text{norm}}(\theta)$ using the uniform-array closed form (from LINEAR ANTENNA ARRAY box):

$$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}, \quad \gamma = kd \cos \theta$$

Need: N , k , d , then form γ , then normalize by peak N^2 .

Solve each unknown:

N : given.

$$N = 5$$

k : wavenumber.

$$k = \frac{2\pi}{\lambda}$$

d : given.

$$d = \frac{3\lambda}{4}$$

γ : combine.

$$\gamma = kd \cos \theta = \frac{2\pi}{\lambda} \cdot \frac{3\lambda}{4} \cos \theta = \frac{3\pi}{2} \cos \theta$$

So $N\gamma/2 = (5/2)(3\pi/2) \cos \theta = (15\pi/4) \cos \theta$ and $\gamma/2 = (3\pi/4) \cos \theta$.

Plug into F_a :

$$F_a(\theta) = \frac{\sin^2((15\pi/4) \cos \theta)}{\sin^2((3\pi/4) \cos \theta)}$$

Peak occurs at $\gamma = 0$ ($\cos \theta = 0$, broadside $\theta = \pi/2$): $F_a^{\text{max}} = N^2 = 25$.

Normalize:

$$F_a^{\text{norm}}(\theta) = \frac{1}{25} \frac{\sin^2((15\pi/4) \cos \theta)}{\sin^2((3\pi/4) \cos \theta)}$$

(b) Half-power beamwidth β (3 pts)

Goal: find β using the uniform-broadside-array HPBW formula:

$$\beta \approx 0.886 \frac{\lambda}{Nd} \quad (\text{rad})$$

Need: N , d , λ .

Plug in:

$$Nd = 5 \cdot \frac{3\lambda}{4} = \frac{15\lambda}{4}$$

$$\beta = 0.886 \cdot \frac{\lambda}{15\lambda/4} = 0.886 \cdot \frac{4}{15} = 0.236 \text{ rad}$$

$$\boxed{\beta = 0.236 \text{ rad} = 13.5^\circ}$$

Cross-check by solving $F_a^{\text{norm}} = 0.5$ numerically: get $\gamma/2 \approx 0.28$ rad, so $\cos \theta = 0.28/(3\pi/4) = 0.119$, $\theta = 83.2^\circ$, offset from broadside (90°) by 6.8° , full HPBW $\approx 13.6^\circ$ ✓.

Sanity: 5 elements give a moderately narrow beam ($\sim 14^\circ$) vs. a single dipole's $\sim 78^\circ$. Adding more elements or wider spacing narrows further (with grating-lobe risk if $d > \lambda$).